Dodona 01 — cluster sanity

go / no-go, no load 2026-10-10 17:05 UTC

Objective

Does the cluster serve a real completion, and does it go THROUGH the control plane rather than past it?

Falsified by: orch_requests_total not moving, or a pool not answering

The promises everything later is scored against

SLO	target	direction
TTFT p95	1.5s	max
turn p95	30s	max
preemptions	0	max
shed rate at target load	0.05	max
concurrent sessions	derived in 02	min

From benchmarking/slos.py. The session target is derived in 02.

Concurrency target not measured yet — run 02-capacity.

1. Is anything listening?

PASS gateway /metrics

PASS gateway /v1/models

PASS prefill /metrics

PASS decode /metrics

>> All four endpoints answer, so the readings below are real.

2. Which pool does which job?

PASS gateway advertises: Qwen/Qwen2.5-7B-Instruct-AWQ

pool	running	waiting	prompt tok	gen tok
prefill	0	0	353,287	130
decode	0	0	353,287	260

prompt tok of 0 on a pool means nothing was ever routed to it.

>> Prefill generated 130 tokens, too many for a max_tokens=1 leg. Check LAB_SPLIT and queue.py.

3. How much room does one pod have?

quantity	value	where it comes from
slice x util	10.80 GiB	HAMi gpumem x --gpu-memory-utilization
weights	5.20 GiB	Qwen2.5-7B AWQ int4, embeddings in fp16
activations + CUDA graphs	1.30 GiB	budget minus weights minus KV
KV cache	4.30 GiB	pod log: Available KV cache memory
seqs per pod at 16k	4.9	pod log: Maximum concurrency
decode pool	4.9	seqs per pod x replicas

PASS a pool of 4.9 is small enough for KV to bind

4.9. That is the scarcity the admission experiments need. The whole card would suggest 73.5 — do not quote that for a slice.

4. Does a request go through the gateway?

```
PASS  orch_requests_total 130 -> 131
PASS  completion in 0.327s
  field    value
  -------  ------------------------
  content  'ok'
  via      local
  latency  0.327s (tunnel included)
```

>> The counter moved and the model answered 'ok'. Guardrails, admission, placement and the hop all ran. via=local means our cluster served it rather than bursting to a provider.

5. Did any KV actually move?

```
PASS  the router RECORDED a hop
  field    value
  -------  ----------------
  src      prefill-0
  dst      decode-0
  req_id   http
  tokens   16
  backend  mooncake
  prefix   802848469d1623df
```

a record, not evidence of bytes

```
  counter                      value
  -------------------------  -------
  connector asked             52,907
  connector returned blocks        0
  tokens arrived as KV             0
  decode's own cache hits    300,416
```

```
FAIL  the connector has delivered at least one block
```

>> The connector was consulted 52,907 times and returned NOTHING. Decode recomputes every prompt. The 300,416 cache hits are decode's own prefix cache serving its own repeats, which is a different thing and is often mistaken for this one. The hop is bookkeeping.

6. Is anything scarce right now?

```
  pool     kv_cache_usage_perc  vs 0.90
  -------  -------------------  -------
  prefill               0.0000  below
  decode                0.0000  below
```

>> Peak KV usage is 0.0% on an idle cluster, so nothing is scarce and the policy has nothing to decide. That is expected here. 02-capacity finds the load at which this crosses 90%, and until it does every admission arm would measure an inert component.

7. Latency on a small request

metric	seconds
n	10
min	0.299
p50	0.333
p95	0.344
max	0.344

>> p50 0.333s, p95 0.344s, spread 0.045s across ten requests, drifting +5% end to end. Stable. The tunnel is inside these numbers, so use make load to quote any.

8. Does decode pay for the prompt?

prompt tokens	p50 seconds
256	0.345
1,024	0.345
4,096	0.478
8,192	0.685
12,288	0.622

PASS all 5 prompt lengths completed
PASS latency grew 1.8x while the prompt grew 48x

>> From 256 to 12,288 prompt tokens (48x), p50 went 0.345s to 0.622s (1.8x). Decode is paying for the prompt, which agrees with section 5: no blocks moved.

9. Grafana

vLLM — KV, running/waiting, TTFT, prefix cache
`http://127.0.0.1:31495/d/dodona-vllm?from=now-30m&to=now`
Gateway — admit, shed, place, latency by stage
`http://127.0.0.1:31495/d/dodona-gateway?from=now-30m&to=now`
Router — pick, sticky, KV hop
`http://127.0.0.1:31495/d/dodona-router?from=now-30m&to=now`
KEDA — scaler value against replicas
`http://127.0.0.1:31495/d/dodona-keda?from=now-30m&to=now`
HAMi — device memory against the slice cap
`http://127.0.0.1:31495/d/dodona-hami?from=now-30m&to=now`
explore: KV usage per pod

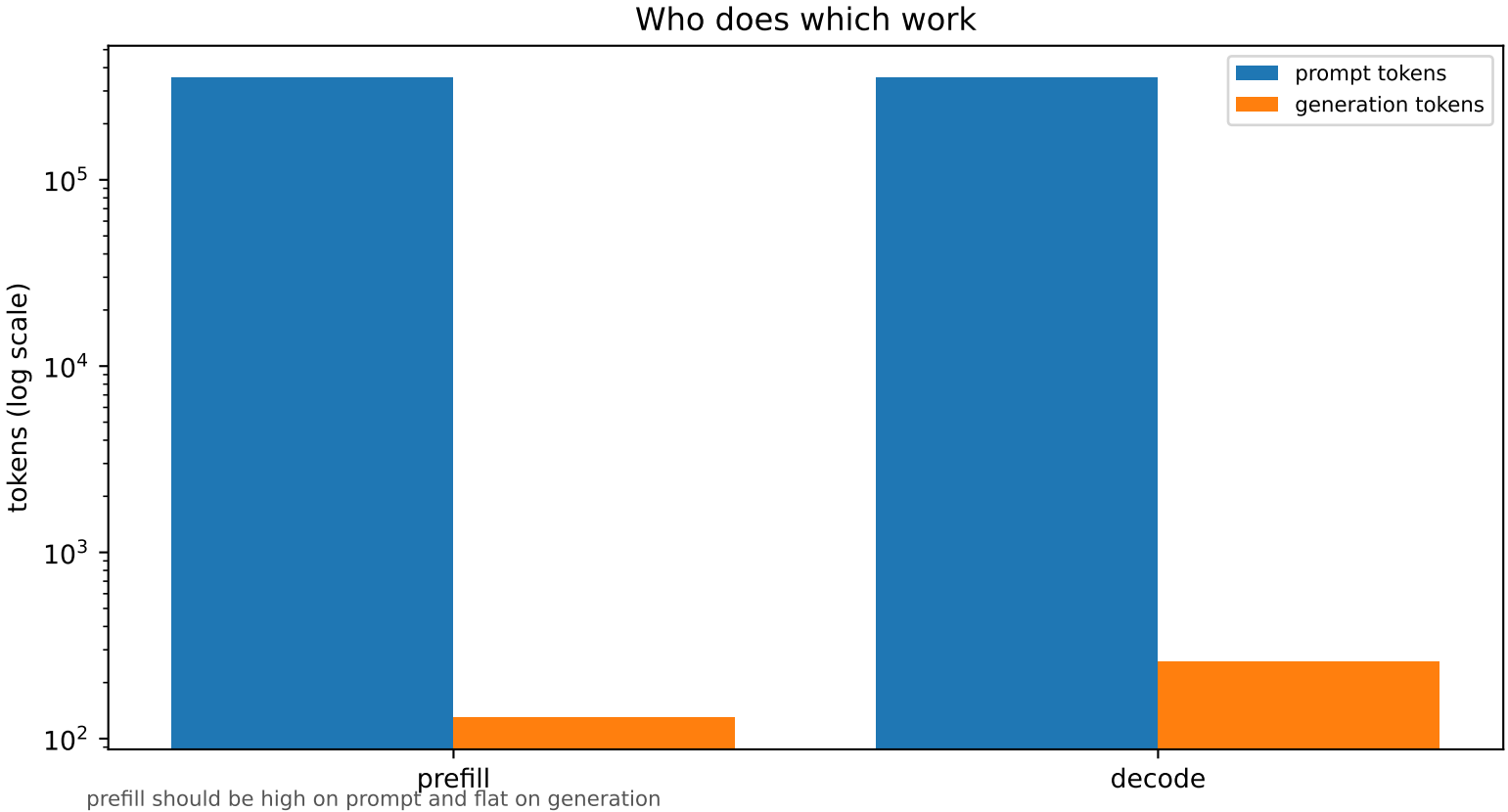
http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22

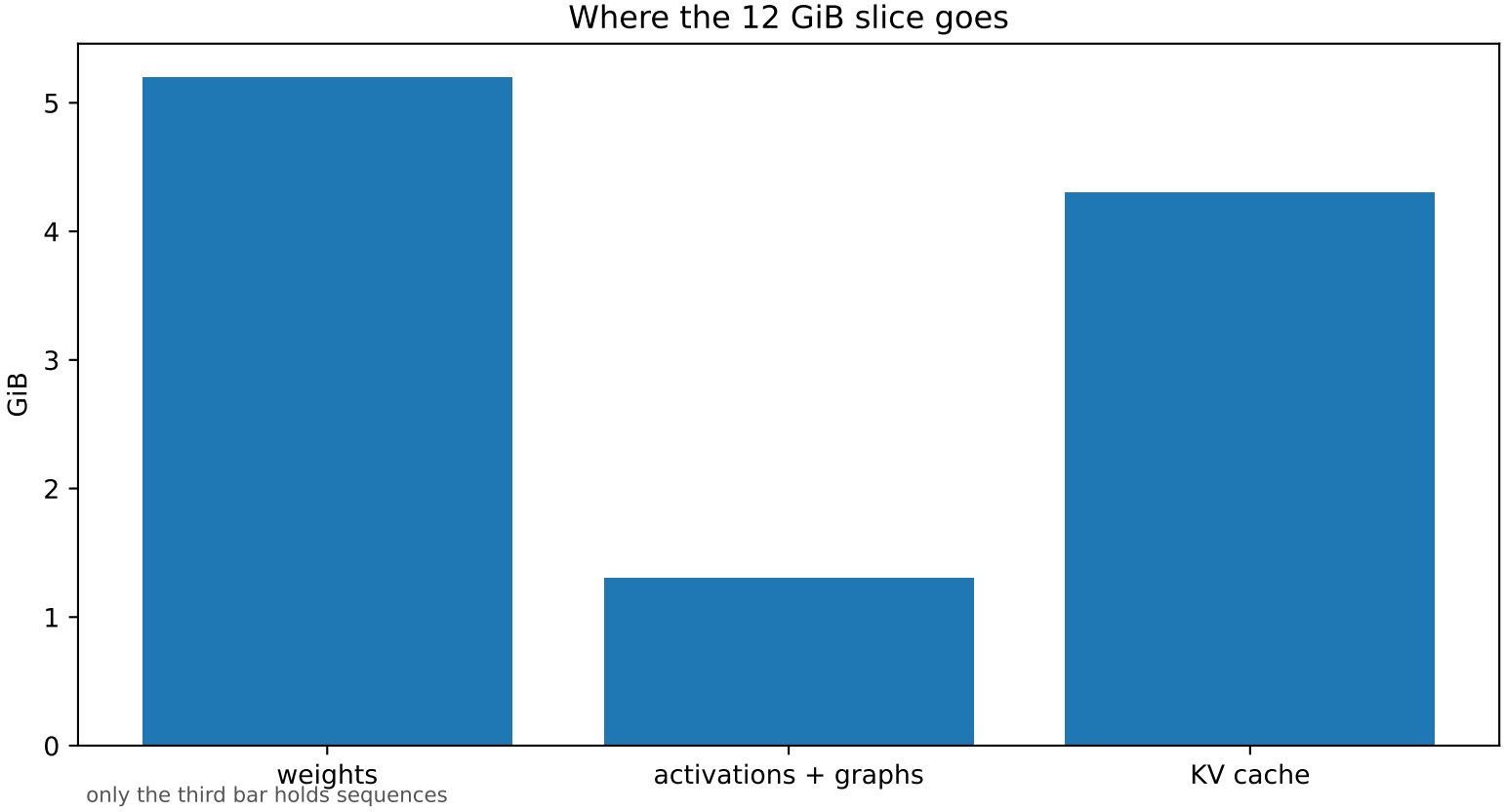
explore: queue depth per pod

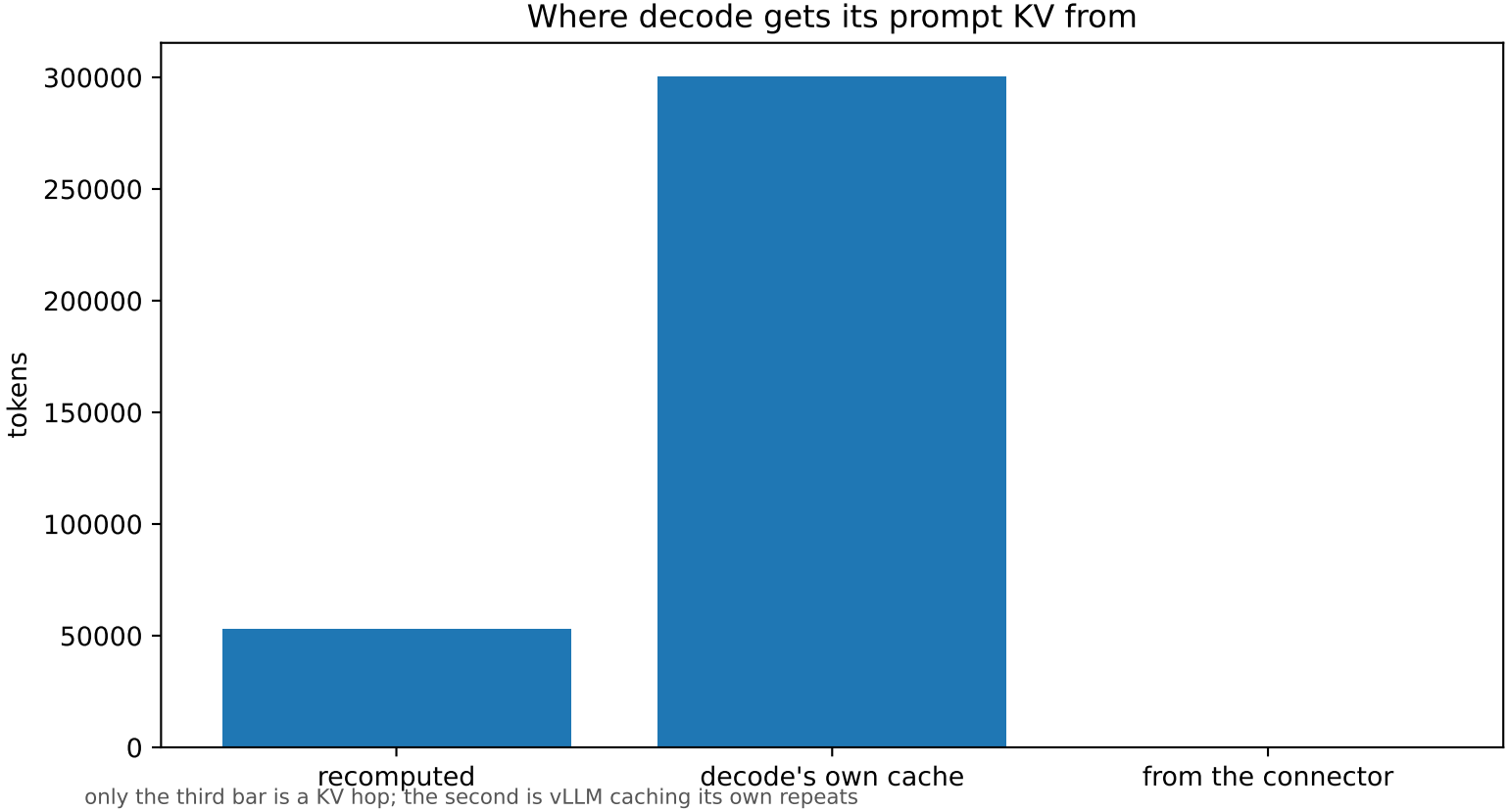
http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22

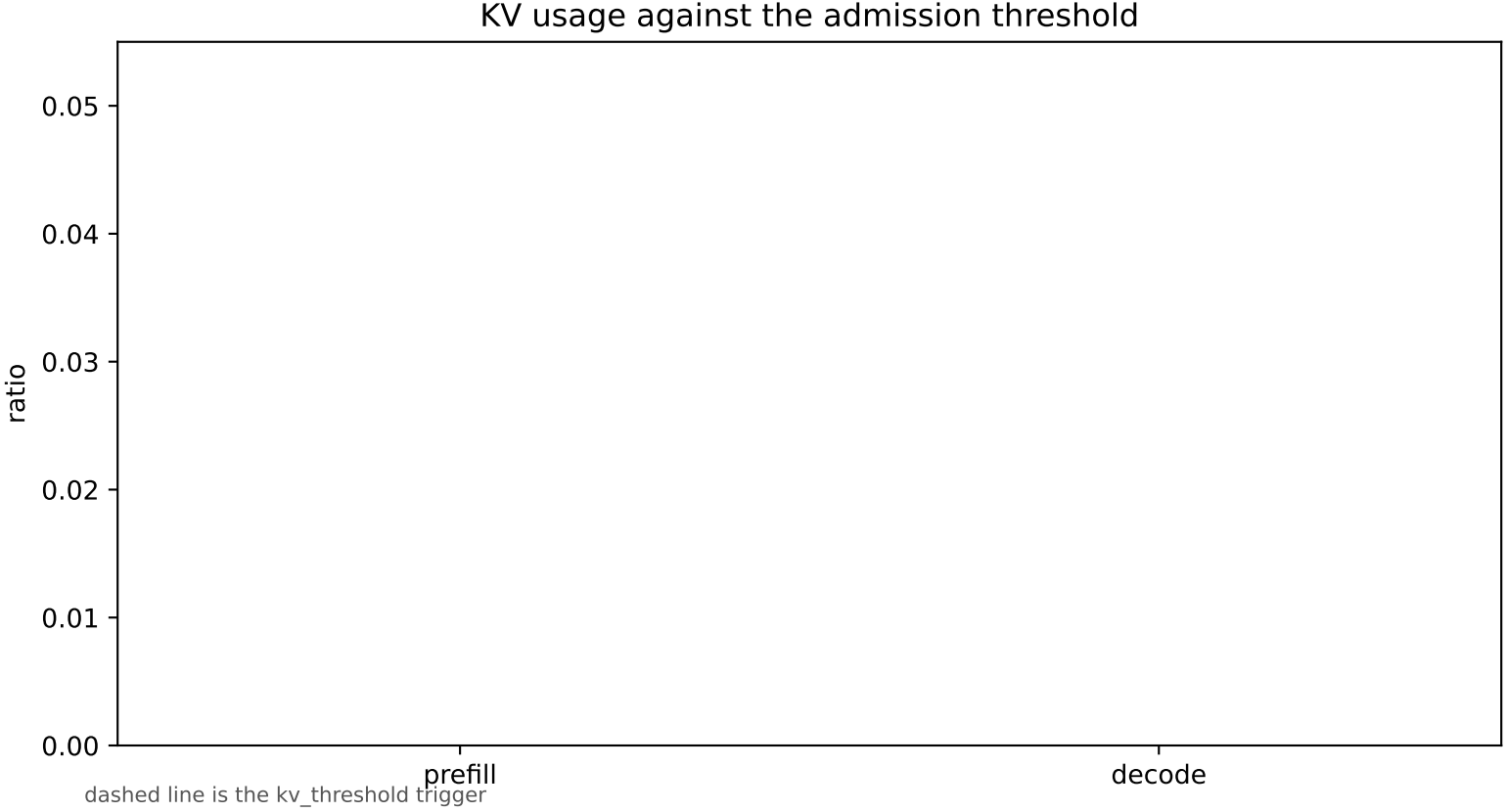
explore: KV actually transferred

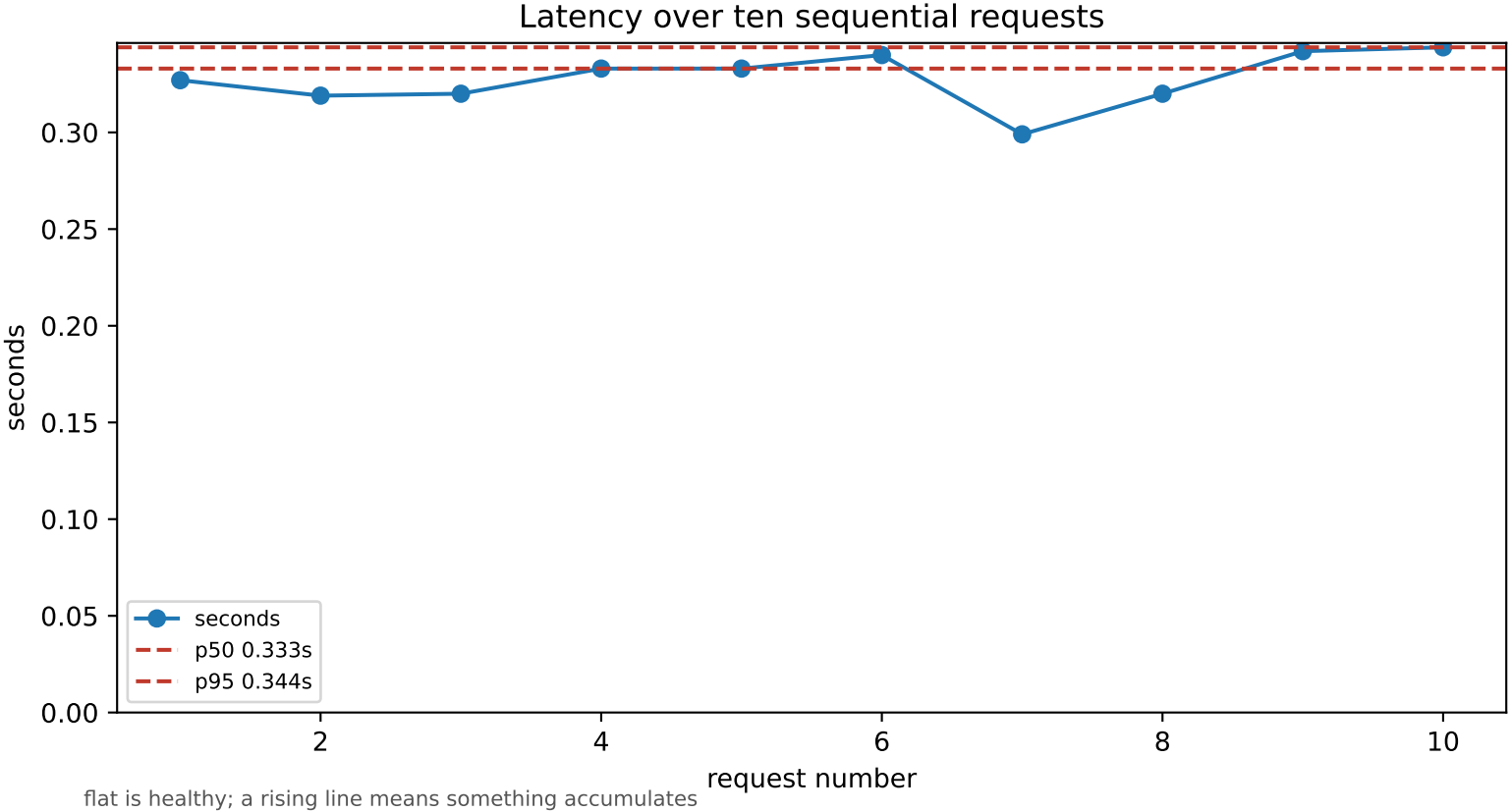
http://127.0.0.1:31495/explore?left=%7B%22datasource%22%3A%20%22Prometheus%22%2C%20%22queries%22%3A%20%5B%7B%22


Who does which work
prompt tokens
generation tokens
tokens (log scale)
10^5
10^4
10^3
10^2
prefill
decode
prefill should be high on prompt and flat on generation

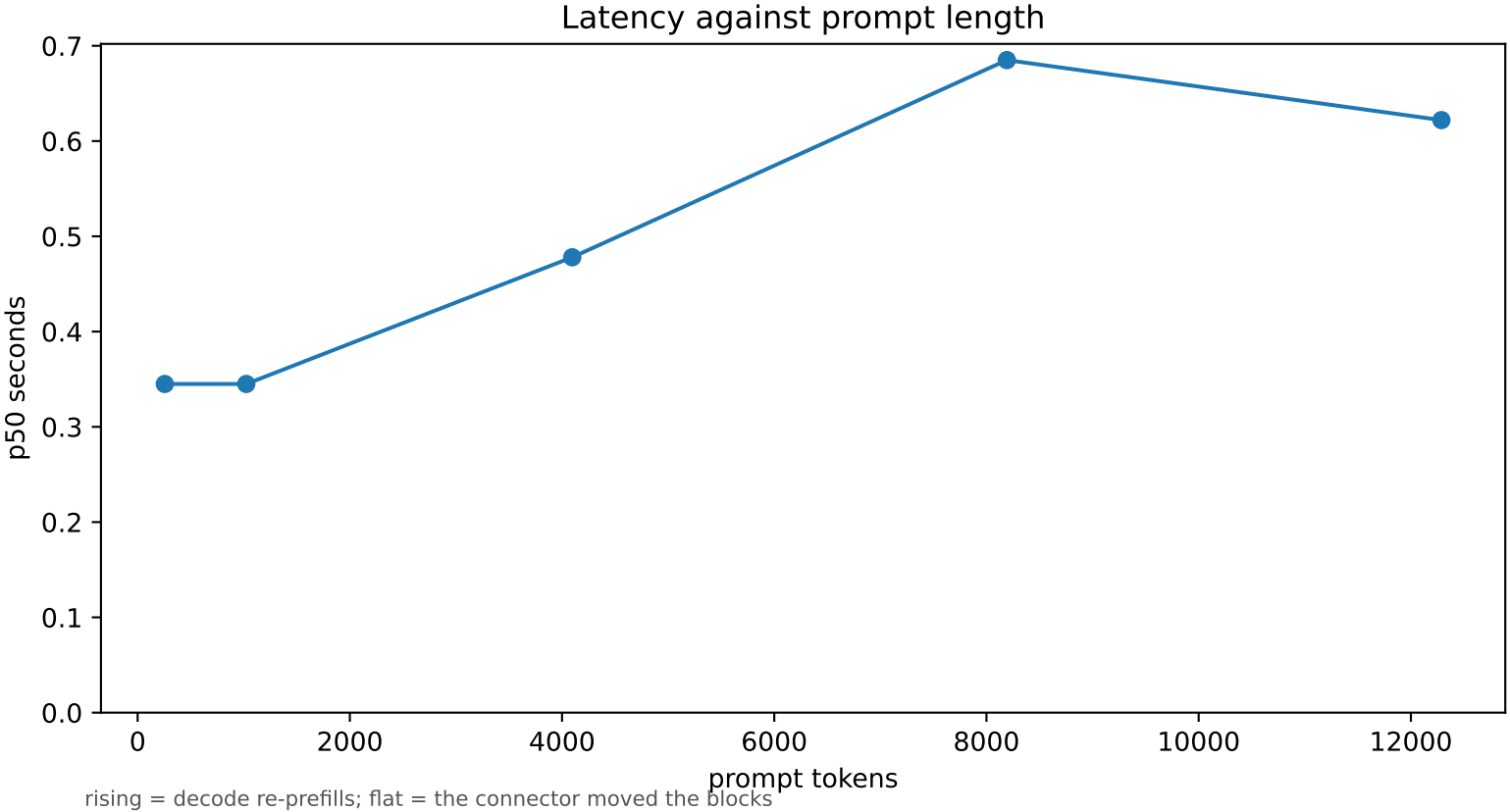

Where the 12 GiB slice goes
GiB
0
1
2
3
4
5
weights
activations + graphs
KV cache
only the third bar holds sequences


Where decode gets its prompt KV from
tokens
300000
250000
200000
150000
100000
50000
0
recomputed
decode's own cache
from the connector
only the third bar is a KV hop; the second is vLLM caching its own repeats


KV usage against the admission threshold
ratio
0.05
0.04
0.03
0.02
0.01
0.00
prefill
decode
dashed line is the kv_threshold trigger



flat is healthy; a rising line means something accumulates


Latency against prompt length
p50 seconds
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
2000
4000
6000
8000
10000
12000
prompt tokens
rising = decode re-prefills; flat = the connector moved the blocks